

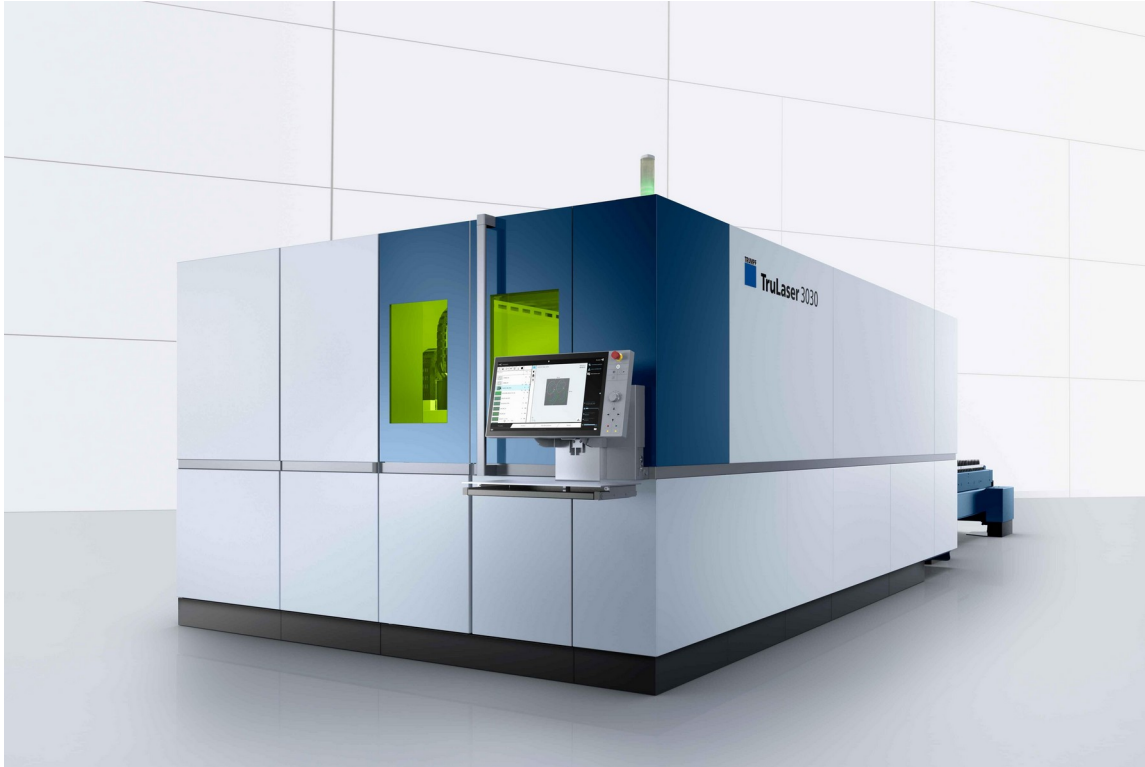
Technical Quotation



ASGCO Manufacturing Incorporated
730 Bangor Road
Nazareth, Pennsylvania 18064
USA

TruLaser 3030 fiber (L95)

Quotation number: 00101508/1 from 01/26/2024
valid to: 02/25/2024



All illustrations are examples. The product may differ from the picture.

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Technical Quotation



ASGCO Manufacturing Incorporated
730 Bangor Road
Nazareth, Pennsylvania 18064
USA

Dear Customer,

Thank you for your interest in TRUMPF technology! In response to your request we are pleased to provide the following quotation.

TRUMPF machines are designed to meet the increasing challenges facing today's manufacturers.

It is most important for us that you benefit from your TRUMPF machine every day and in every step you are using it. Using TRUMPF accessories and services is the key to attain the full potential of your new TRUMPF machine and will ensure a maximum uptime, low operating cost and maintain its high capital value. We bundle these products under the category TruServices. TRUMPF knows it is our duty to develop and continuously improve TruServices to ensure your competitive advantage for the whole time you use your machine. This is why we include superior services with your new machine such as our TRUMPF warranty, consumables, training, installation and data-driven services for monitoring an optimal performance.

As the largest laser manufacturer in the world, and the largest manufacturer of fabrication equipment in the United States, TRUMPF is positioned to offer innovative machines of very high reliability and quality. TRUMPF has been in business for over 100 years and spends approximately 10% of its revenues on research and develop programs that result in continued enhancements of existing products along with innovative new technologies to solve tomorrow's fabrication needs.

While our unique fabrication machines set themselves apart from our competitors' products through their design and productivity, the company also distinguishes itself through dedication to customer training and support. More than 3,000 students pass through our US customer training center each year, learning to program, to operate and maintain their TRUMPF products. We are also constantly optimizing our technical service group, keeping ahead of our install base. Our dedication to continuous innovation and customer support has helped the TRUMPF Group to exceed \$5.5 billion in sales worldwide.

If you have not already had the opportunity to experience our 435,000 square foot manufacturing facility in Farmington, CT, we would like to offer you a visit. Meeting the people of TRUMPF while getting a firsthand look at our facility, our products and our lean manufacturing operations will open your eyes to the TRUMPF philosophy. And if Connecticut is not your backyard our technology is also showcased in our SmartFactory in Chicago, IL and in our Technology Center in Costa Mesa, CA. Contact your local TRUMPF sales person to arrange a visit.

Thank you once again for considering TRUMPF!

Burke Doar
Executive Vice President

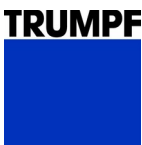


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Technical Quotation



1. Quotation

Pos	Description	Amount	Position price
1000	TruLaser 3030 fiber (L95) - Machine - Working range: 3000 × 1500 mm - Box frame - Longitudinal conveyor belt - Motion unit for high-precision processing - Gantry drive in combination with linear direct drives - Process cooler for machine and laser (water/air) - Work area illumination - Pilot laser - Spraying equipment - Automatic nozzle cleaning - Integrated electrical cabinets - Integrated cabinet - Closed beam guidance via laser light cable (LLK) - Machine status light - Transport and installation device - TRUMPF laser - Excitation using pump diodes - Laser power control - Cutting unit - Single cutting head strategy - Universal cutting unit with adaptive lens system - Protective glass - Online condition checking of protective glass - Automatic nozzle centering - Control - Quick post-production - SINUMERIK control - Ergonomic control panel with 21.5" wide-screen multi-touch color display - Integrated technology data - Auto-shutdown - Programmable cutting gas/pressure selection - Microweld - Remote support interface	1 PCE	755,000.00 USD

Technical Quotation



- Production plan
- Multi-sheet processing
- Condition Guide
- Cutting Guide
- Functions
 - AdjustLine
 - ControlLine
 - FastLine
 - FlyLine
 - FocusLine
 - NitroLine
 - PierceLine
 - PlasmaLine
 - Copper cutting package
 - Brass cutting package
- Data transmission
 - USB interface
 - RJ45 network interface
 - Central Link
 - Online Update Manager
- Safety
 - CE marking and EC Machinery Directive
 - Light barriers
 - Multi-chamber compact dust extractor
 - Enclosed protective housing with certified door viewing panel, right
- Advanced standard configuration
 - External solid-state laser TruDisk 4001 with one laser output
 - Loading station for 90 mm stack height
 - Mild steel rake (prong spacing 67 mm, replaceable prongs)
 - Finished part pallet

2000	Standard TRUMPF TruServices	1 PCE	0.00 USD
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Warranty:

One-Year, Unlimited Hours

Training:

2 students for a 4 day operator course at TRUMPF's facility in Farmington, CT

2 students for a 2 day advanced application training course at TRUMPF's facility in Farmington, CT

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1 day of operators review after installation

Transparency Agreement:

Machine will be connected to our Condition Monitoring Platform including the following benefits:

Data-driven diagnostics: Faster and more reliable problem solving based on machine data

Detection and prevention of machine failures and unplanned downtimes

Condition Report via MyTRUMPF: Machine performance data going back 7 days, so that you can analyze them and take the necessary steps to improve your operation

Valid for five years

Installation:

Installation, Start-up and Instruction

Standard Acceptance:

Machine acceptance will consist of producing TRUMPF's standard acceptance part which checks all aspects of machine performance against quoted specifications. This acceptance test will be performed at the customer's location, if requested. Any additional tests must be mutually agreed upon and specified in the purchase order.

3000	External solid-state laser TruDisk 6001 with one laser output	1 PCE	100,000.00 USD
	• Includes BrightLine fiber and Highspeed cutting package		
4000	External solid-state laser TruDisk 10001 with one laser output	1 PCE	300,000.00 USD
	• Includes BrightLine fiber and Highspeed cutting package		
5000	External solid-state laser TruDisk 12001 with one laser output	1 PCE	400,000.00 USD
	• Includes BrightLine fiber and Highspeed cutting package		
6000	External solid-state laser TruDisk 24001 with one laser output	1 PCE	810,000.00 USD
	• Includes BrightLine fiber, Highspeed cutting package and universal cutting unit with fully adaptive lens system		
7000	Power by the Hour program	1 PCE	-200,000.00 USD
8000	Nozzle changer	1 PCE	25,000.00 USD
	• 21 nozzle positions		

Technical Quotation



9000	DetectLine	1 PCE	12,000.00 USD
	Precise sheet position detection (including automatic focal measurement)		
10000	Drop&Cut	1 PCE	6,000.00 USD
	Only in combination with integrated camera 1 or integrated cameras 1 and 2		
11000	Integrated camera 1 for process viewing	1 PCE	4,000.00 USD
	View of the front section of the work area (about 2.5 m)		
12000	Integrated camera 2 for process viewing	1 PCE	4,000.00 USD
	- View of the remaining section of the work area (from 6 m format view up to 4.5 m) - Only in combination with integrated camera 1		
13000	Smart Collision Prevention / Purchase	1 PCE	10,500.00 USD
	- Only in combination with TruTops Laser or TruTops Boost - Unlimited time license		
14000	Dot Matrix Code / Purchase	1 PCE	3,000.00 USD
	- Part identification with Dot Matrix Code - Additional function: rapid and simple application of free text with minimal programming effort - Requirement TruLaser: TruTops Boost version 8 or above - Requirement TruLaser Tube: Programming Tube version 1 or higher - Unlimited time license		
15000	Highspeed Eco cutting package	1 PCE	0.00 USD
	High-speed cutting with low gas consumption		
16000	CoolLine with water container	1 PCE	15,000.00 USD
	Water supply is done using a water container		
17000	FlexLine	1 PCE	15,000.00 USD
	Quality process for processing diverse structural steel grades and surface qualities		
18000	Compressed-air cutting	1 PCE	0.00 USD
19000	EdgeLine Bevel / Purchase	1 PCE	25,000.00 USD
	Unlimited time license		
20000	Smart Rerun / Purchase	1 PCE	7,500.00 USD
	Unlimited time license		

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21000	Titanium cutting package - Includes cutting data N2 and argon - Contains additional cutting gas connection - Includes compact dust extractor with fire suppression	1 PCE	5,000.00 USD
22000	Additional cutting gas connection	1 PCE	4,500.00 USD
23000	Gas mix cutting package - Includes cutting data for gas mixture N2/O2 - Includes e-learning course on adapting cutting data from the cutting package gas mix to the customer's individual material (mat. no. 2764083) - Includes the option "Additional cutting gas connection" (1425430)	1 PCE	9,500.00 USD
24000	CO2 extinguisher system for compact dust extractor	1 PCE	12,000.00 USD
25000	Cutting package for thick sheets	1 PCE	0.00 USD
26000	Clamps (2x for one pallet)	1 PCE	1,000.00 USD
27000	Clamps (4x for two pallets)	1 PCE	2,000.00 USD
28000	Cleaning brush	1 PCE	4,500.00 USD
29000	TruLaser L95 Convenience Package Convenience Package for each machine will include: - Electric Rollup Roof - Longitudinal conveyor belt - Ergonomic Control Panel with 21.5" wide screen multi touch color display - Foldable front door with left and right viewing windows.	1 PCE	0.00 USD
30000	Transverse conveyor belt with tipping container (to the left of the machine)	1 PCE	10,000.00 USD
31000	Transverse conveyor belt with tipping container (to the right of the machine)	1 PCE	10,000.00 USD
32000	Machine ambient temperature up to 45 °C Dependent on relative humidity, see installation conditions	1 PCE	0.00 USD
33000	MobileControl app	1 PCE	2,500.00 USD

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- Requirement: Wireless Operation Point

34000	Wireless Operation Point	1 PCE	2,500.00 USD
35000	Additional 10-m LLK extension between the laser and machine • Up to TruDisk 4001: Max. 20 m in addition • From TruFiber 3001/TruDisk 6001 on: max. 10 m additional	1 PCE	5,000.00 USD
36000	Right-hand installation of the machine	1 PCE	5,000.00 USD
37000	Mezzanine version of machine • Mezzanine structure not included	1 PCE	10,000.00 USD
38000	Sorting Guide • Scope of delivery: IPC with software, camera and monitor • Functionality and detection precision depend on customer lighting (see installation conditions)	1 PCE	35,000.00 USD
39000	Tube cutting unit including four standard and four reversible jaws, support system inside • Scope of delivery: Rotary unit with dynamic torque direct drive, 4-jaw collet chuck for clamping round and square tubes, exhaust system, NC stop for quicker stopping of inserted tubes and profiles, guide rail for support, 1 x internal support • Includes electrical rolling roof, door viewing panel (left) for extended visibility, as well as roller cabinet for accessories Integrated cabinet omitted	1 PCE	85,000.00 USD
40000	External support max. D 20-165 mm (for pipes up to 6 m, includes two step rollers) • Includes two step rollers	1 PCE	10,000.00 USD
41000	Additional internal support D 20-370 mm	1 PCE	6,500.00 USD
42000	LiftMaster Compact retrofit capability • With extended pallet removal unit • Retrofit capability for three years after delivery of machine	1 PCE	0.00 USD

Automation

43000	LoadMaster 1530 • Basic unit • Type 1530: sheet sizes/thicknesses 3048 × 1524 mm/loading 25 mm	1 PCE	79,000.00 USD
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Technical Quotation



	• Loading • Loading station • Fanning magnets		
44000	Peeling unit	1 PCE	0.00 USD
45000	Second light barrier circuit • Necessary if the safety area is interrupted by a wall or similar obstruction • Acknowledgement occurs together with the first light barrier circuit	1 PCE	0.00 USD
46000	Front dual loading cart B1 • Includes storage interface, fanning magnets, operating panel and manual cart control • Loading station (340959) removed from standard scope of delivery	1 PCE	90,000.00 USD
47000	Front loading cart B1 • With fanning magnets and manual cart control • Loading station (340959) removed from standard scope of delivery	1 PCE	45,000.00 USD
48000	Installation on opposite side from electrical cabinet	1 PCE	0.00 USD
49000	LiftMaster 1530 (AL35) • Basic unit • Type 1530: sheet sizes/thicknesses 3048 × 1524 mm/loading 25 mm/unloading 25 mm • Symmetrical installation with loading on the left or right • Loading • Double sheet detection • Suction frame • Peeling unit • Separation of thin sheet	1 PCE	210,000.00 USD
50000	Loading station • For stack heights of up to 200 mm	1 PCE	0.00 USD
51000	Unloading platform	1 PCE	0.00 USD
52000	Rake with belt prongs for low-scratch, positioning accurate unloading (prong spacing 67 mm)	1 PCE	20,000.00 USD

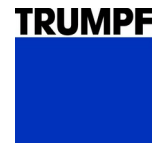
Technical Quotation



- Available up to sheet thickness $s = 15 \text{ mm}$
- Extra charge
- Mild steel rake (prong spacing 67 mm) from standard scope of delivery not required

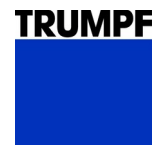
53000	Rake with belt prongs for low-scratch, position accurate unloading (prong spacing 33.5 mm)	1 PCE	30,000.00 USD
	• Available up to sheet thickness $s = 15 \text{ mm}$ • Extra charge • Mild steel rake (prong spacing 67 mm) from standard scope of delivery not required		
54000	Part separation from 600 × 600 mm	1 PCE	20,000.00 USD
55000	Integrated sheet alignment	1 PCE	0.00 USD
56000	Third position	1 PCE	0.00 USD
	• Necessary if loading and unloading do not take place in the same position		
57000	Base for LiftMaster	1 PCE	0.00 USD
	• Design according to layout • Necessary for storage connection with dual cart		
58000	Storage interface	1 PCE	0.00 USD
59000	LiftMaster Compact 1530 (AL34)	1 PCE	218,000.00 USD
	• Basic unit • Type 1530: sheet sizes/thicknesses 3048 × 1524 mm/loading 25 mm/unloading 25 mm • Two gantry stations • Extended pallet removal unit • Storage interface • Touch start post • Loading • Double sheet detection • Synchronous loader including separation of thin sheet • Peeling unit • Cleaning brush for slats		
60000	Thin sheet handling	1 PCE	15,000.00 USD
	• Mark-free depositing of sheet metal onto slats • Available for sheet thicknesses $s = 0.4 - 2.0 \text{ mm}$		
61000	Load station for expanded stack height	1 PCE	23,000.00 USD

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	- For stack heights of up to 300 mm - Loading station from standard scope of delivery not required		
62000	Loading station for expanded stack height in TruStore 3030	1 PCE	23,000.00 USD
	- For stack heights of up to 260 mm - Loading station from standard scope of delivery not required		
63000	Unloading platform	1 PCE	0.00 USD
64000	Rake with belt prongs for low-scratch, positioning accurate unloading (prong spacing 67 mm)	1 PCE	20,000.00 USD
	- Available up to sheet thickness s = 15 mm - Extra charge - Mild steel rake (prong spacing 67 mm) from standard scope of delivery not required		
65000	Rake with belt prongs for low-scratch, position accurate unloading (prong spacing 33.5 mm)	1 PCE	30,000.00 USD
	- Available up to sheet thickness s = 15 mm - Extra charge - Mild steel rake (prong spacing 67 mm) from standard scope of delivery not required		
66000	PartMaster	1 PCE	76,000.00 USD
	- Available up to sheet thickness s = 12.7 mm - Includes scrap container under the part chute		
67000	PartMaster at a gantry station	1 PCE	96,000.00 USD
	Available up to sheet thickness s = 12.7 mm		
68000	PartMaster integration TruStore	1 PCE	76,000.00 USD
	Includes PartMaster (2464203), specific storage compartment extension and pallet handling extension (including 4 long-sided grippers) (2464205)		
69000	Additional finished part pallet	1 PCE	5,000.00 USD
70000	Part separation from 150 × 150 mm	1 PCE	50,000.00 USD
	- Available up to sheet thickness s = 8 mm - Includes sensor system for separation checking - When used with LiftMaster Compact, the scrap skeletons must be deposited separately from the finished parts		
71000	Linear version	1 PCE	37,000.00 USD

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additional gantry station

72000	Standing electrical cabinet	1 PCE	8,000.00 USD
	• Is required if it is not possible to mount the control cabinet to the gantry station • Includes rack for electrical cabinet and cable sets		

Automation Carts

73000	Loading cart	1 PCE	45,000.00 USD
	• Contains cart interface • For automation with format 1530: At least 9 m rail length is recommended, maximum 3 t carrying capacity • For automation with format 2040: At least 11 m rail length is recommended, maximum 5 t carrying capacity • For automation with formats 2060/2560: At least 15 m rail length is recommended, maximum 10 t carrying capacity		

74000	Unloading cart	1 PCE	45,000.00 USD
	• Contains cart interface • For automation with format 1530: At least 9 m rail length is recommended, maximum 3 t carrying capacity • For automation with format 2040: At least 11 m rail length is recommended, maximum 5 t carrying capacity • For automation with formats 2060/2560: At least 15 m rail length is recommended, maximum 10 t carrying capacity		

75000	Dual loading cart	1 PCE	90,000.00 USD
	• Contains cart interface • For automation with format 1530: At least 9 m rail length is recommended, maximum 3 t carrying capacity • For automation with format 2040: At least 11 m rail length is recommended, maximum 5 t carrying capacity • For automation with formats 2060/2560: At least 15 m rail length is recommended, maximum 10 t carrying capacity		

76000	Double unloading cart	1 PCE	90,000.00 USD
	• Contains cart interface • For automation with format 1530: At least 9 m rail length is recommended, maximum 3 t carrying capacity • For automation with format 2040: At least 11 m rail length is recommended, maximum 5 t carrying capacity • For automation with formats 2060/2560: At least 15 m rail length is recommended, maximum 10 t carrying capacity		

77000	Double loading / unloading cart	1 PCE	90,000.00 USD
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Technical Quotation



- With cart interface
- For automation with format 1530: At least 9 m rail length is recommended, maximum 3 t carrying capacity
- For automation with format 2040: At least 11 m rail length is recommended, maximum 5 t carrying capacity
- For automation with formats 2060/2560: At least 15 m rail length is recommended, maximum 10 t carrying capacity

78000	Travel rails, surface-mounted	1 m	0.00 USD
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79000	Travel rails, embedded	1 m	0.00 USD
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- Recess depth: 45 mm

80000	Front loading cart (with scissor lift) for expanded stack height	1 PCE	50,000.00 USD
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- For automation with format 1530: 300 mm stack height at 3 t carrying capacity
- For automation with format 2040: 300 mm stack height at 5 t carrying capacity
- Loading station (340959) removed from standard scope of delivery

TRUMPF TruServices

81000	Extended Warranty + 2 PM Mission	1 PCE	50,000.00 USD
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Warranty extension with 2 preventative maintenance missions per year. (1 year max)

82000	Additional Preventive Maintenance visit	1 PCE	8,000.00 USD
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83000	Service Agreement Remote	1 PCE	750.00 USD
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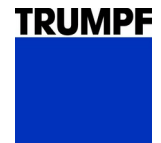
Includes technical remote support provided by an in-house service technician. Unlimited use of TRUMPF Visual Assistance on Smart Glasses or any mobile device. Technical Guides are available upon use of Service App.

Condition Monitoring: option, which identifies anomalies using machine data analysis, includes:

- Proactive expert contact by phone
- Proactive recommendations for action by mail or service app
- Proactive problem solving by TRUMPF service technicians

Condition Monitoring recommended for compatible machines at an

Technical Quotation



additional fee.

84000	Service Agreement Maintenance	1 PCE	17,000.00 USD
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Includes 2 preventative maintenance missions per year for TruLaser, TruPunch, TruMatic, TruLaser Tube, TruLaser Center and 1 preventative maintenance mission per year for TruBend , a 10% discount on consumables, 10% discount on replacement parts and 24/7 technical remote support via phone or TRUMPF Visual Assistance platform (4 add. years max)

Condition Monitoring required for compatible machines for an additional fee.

Customer can opt out of condition monitoring.

85000	Service Agreement Maintenance + Onsite Repair Coverage	1 PCE	29,000.00 USD
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Includes benefits of Maintenance Service Agreement and unlimited repair missions.

Condition Monitoring: option, which identifies anomalies using machine data analysis, includes:

- Proactive expert contact by phone
- Proactive recommendations for action by mail or service app
- Proactive problem solving by TRUMPF service technicians

Condition Monitoring recommended for compatible machines at an additional fee.

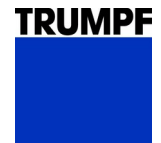
86000	Service Agreement Performance	1 PCE	50,000.00 USD
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Error correction by telephone • Remote support: Online remote diagnostics and error correction • Technical Guides • Regular maintenance performed to specifications of the maintenance checklist • Service missions for repairs • Service Parts for repairs (4 add. years max, 3 add. years after Warranty Extension)

Condition Monitoring: option, which identifies anomalies using machine data analysis, includes:

- Proactive expert contact by phone
- Proactive recommendations for action by mail or service app
- Proactive problem solving by TRUMPF service

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technicians

Condition Monitoring recommended for compatible machines at an additional fee.

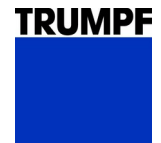
87000	Smart Glasses	1 PCE	2,750.00 USD
We also encourage you to take advantage of our Augmented Reality technology (ART) that provides a virtual means of supporting you faster and more practical in an industrial environment. Instead of waiting for a technician in case of a machine incident, you will receive instructions via smart glasses, so that you can perform a repair quicker yourself. Therefore, we included unlimited augmented reality sessions in all of our service agreements. (Hardware only, but must be purchased with an Agreement)			
88000	Operator Training, at TRUMPF in Farmington, CT	1 PCE	2,500.00 USD
89000	Advanced apps training 2days a TRUMPF	1 PCE	1,600.00 USD
90000	Sheetmetal Design Class, 3 days @ TRUMPF	1 PCE	4,000.00 USD
1 seat included, \$3,500 per additional seat In this course, your employees will learn how to streamline the design process and make full use of sheet metal knowledge to design better parts and replace and/or make improvements to current manufacturing			
91000	2D Laser Applications Boot Camp	1 PCE	5,000.00 USD
2 days @ Customer Site (incl. travel and lodging costs) Customers will receive the following benefits from this training course: Increase programmer and operator knowledge of machine functions and capabilities. Boost overall machine productivity Improve cut part quality.			
92000	Laser Maintenance Level 1	1 PCE	3,200.00 USD
93000	Laser Maintenance Level 2	1 PCE	3,200.00 USD
94000	Maintenance Training at Customer (+exp.)	1 PCE	10,000.00 USD
95000	Yearly Consumable Parts Standard Kit	1 PCE	2,915.00 USD
Standard Ceramics (4 pce) Protective Glass D34 (15 pce) - lower wattage OR Protective Glass D30 (25 pce) - 8kW and higher EAA nozzles (11pks any size)			

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96000	Slim Nozzles Add-on	1 PCE	321.00 USD
	For specific cutting processes, we offer additional packages of nozzles that can only be purchased in addition to the standard package Type of Material Diameter [mm] Quantity EAB (Slim Nozzle) 0.8 5 EAB (Slim Nozzle) 1.0 5 EAB (Slim Nozzle) 1.2 5 EAB (Slim Nozzle) 1.4 5 EAB (Slim Nozzle) 1.7 5 EAB (Slim Nozzle) 2.0 5 EAB (Slim Nozzle) 2.3 5 EAB (Slim Nozzle) 2.7 5 Select 5 pcs of any size Each pack incl. 5 nozzles EAB (Slim): The slim nozzle allows for closer nesting on the sheet metal thereby saving up to 10% on material		
97000	BrightLine Add-on	1 PCE	432.00 USD
	5 pieces of any size BrightLine nozzles		
98000	Performance and Highspeed Nozzles Add-on	1 PCE	659.00 USD
	3 packs of any size Performance and High Speed nozzles		
99000	CoolLine Add-on	1 PCE	1,121.00 USD
	7 pieces of any size CoolLine nozzles & 1 universal ceramic		
100000	Highspeed Eco Add-On	1 PCE	787.00 USD
	1 High Speed Eco EAV nozzle 2 packs of EAV inserts 2 packs of EAV sleeves		
101000	Oseon Analytics (Complete Package - US Standard) - Valid for new SOFTWARE customer only	1 PCE	8,500.00 USD
	• One user license • Implementation of one TRUMPF machine • Software maintenance for 12 months (for new software customer only) or alignment the software maintenance with the current license(s) (existing software customer)		
102000	TruTops Boost Software bundle (Standalone Machine) - Valid for new TruTops Boost customer only	1 PCE	28,800.00 USD
	All in-one design and programming solution that works consistently from 3D modeling to NC generation across different technologies. The integrated software offers connected intelligence for the ideal interaction between software and machine. - One TruTops Boost core (floating license) including one Machine Package - Training seat for one student		

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	- Software maintenance contract for a 12-month period - Telephone & remote support and software updates		
103000	TruTops Boost Software bundle (Machine w/Automation) - Valid for new TruTops Boost customer only	1 PCE	30,300.00 USD
	All in-one design and programming solution that works consistently from 3D modeling to NC generation across different technologies. The integrated software offers connected intelligence for the ideal interaction between software and machine. - One TruTops Boost core (floating license) including one Machine Package - Training seat for one student - Software maintenance contract for a 12-month period - Telephone & remote support and software updates		
104000	TruTops Boost Standard Machine Package - Existing TruTops Boost license(s) as well as valid software maintenance is required	1 PCE	11,500.00 USD
	This option permits programming of the existing machine model quoted. Training not included		
105000	TruTops Boost Automation Machine Package - Existing TruTops Boost license(s) as well as valid software maintenance is required	1 PCE	13,000.00 USD
	This option permits programming of the existing machine model quoted. Training not included		
106000	Additional core (floating license) of TruTops Boost - Existing TruTops Boost license(s) as well as valid software maintenance is required	1 PCE	11,000.00 USD
	Machine package(s) from the first core included. Current Software maintenance contract required / follows first core validity. Training not included		
107000	Data Exchange Package for 3D CAD models - Existing TruTops Boost license(s) as well as valid software maintenance is required	1 PCE	2,700.00 USD
	Read and write: VDA-FS (VDA), Parasolid (X_T, X_B, XMT_TXT, XMT_BIN) Read: Inventor (IPT, IAM), NX (PRT), Pro/E (PRT, ASM, XPR, XAS), SolidEdge (par, psm, asm), Solidworks (SLDPRT, SLDASM)		
108000	Assembly Importer - Existing TruTops Boost license(s) as well as valid software maintenance is required	1 PCE	2,700.00 USD
	Automated dissolving of assemblies in single parts in all supported file formats and automatic transfer into HomeZone (parts		

Technical Quotation



management)

109000	Interface Package Catia V5 - Existing TruTops Boost license(s) as well as valid software maintenance is required	1 PCE	4,500.00 USD
	Supports direct import and export of the file format CATIA V5 (MODEL, EXP, CATpart, CATproduct, CGR)		
110000	TruTops Boost Programming Training	1 PCE	2,800.00 USD
111000	TruTops Boost Design Programming (webinar)	1 PCE	2,400.00 USD
	This course is designed for customers who wish to learn design in TruTops Boost without having to travel to TRUMPF. The goal of the course is to provide students with the core knowledge and skills to successfully draw 2D or 3D parts.		
112000	Either switch from TruTops Classic to Boost or Metamotion software	1 PCE	0.00 USD
	Please contact softwaresales@trumpf.com or ask your Sales Representative how to receive beneficial switch-over (update) pricing from TruTops Classic to TruTops Boost, as well as Metamotion software quote		

If a service agreement is included: Service agreement will start after the warranty period of the machine expires.

2. Terms and conditions

Terms of payment

Payment Terms:

20% deposit due with purchase order
40% prior to shipment
40% Net 10 days from Installation Completion

TRUMPF, INC. TERMS AND CONDITIONS OF SALE FOR MACHINE TOOL EQUIPMENT AND AUTOMATION

This Agreement governs the sale of commercial off-the-shelf and commercial items products from Seller's product lines (hereinafter referred to as "Goods" or "Equipment"). Seller makes all quotations and accepts orders for such Goods/Equipment only on the terms and conditions stated herein ("Terms of Sale"):

1. ORDER ACCEPTANCE:

All orders are subject to acceptance only at Seller's facility in Farmington, Connecticut. These Terms of Sale shall be deemed accepted by Buyer upon Seller's receipt of Purchase Order from Buyer. No condition stated by Buyer shall be binding upon Seller if in conflict with, inconsistent with or in addition to the Terms of Sale, unless expressly accepted in a writing signed by Seller. In the event of conflict or differences in the terms or conditions of Buyer's Purchase Order and the Terms of Sale herein, the Terms of Sale shall govern.

2. PRICES:

All prices are: (a) Seller's current prices and are subject to change without notice at any time prior to acceptance of Buyer's order; (b) subject to all federal, state and local taxes upon the production, sale or shipment of the Goods sold hereunder, now or hereafter becoming effective, and if not included in the invoice, such amount may be invoiced later, and Buyer shall pay all such taxes.

3. SHIPMENTS:

(A) Domestic: All orders are FCA Seller's plant in Farmington, Connecticut (or such warehousing facilities as Seller may establish) Incoterms 2020.

(B) International: All prices are in US Dollars and are, at Seller's election, FCA Farmington CT (or other warehousing facilities as Seller may establish) or DPU Port of Entry Incoterms 2020.

(C) All freight charges, insurance premiums, duties and taxes are the responsibility of the Buyer.

4. PAYMENT:

All payment terms are subject to Seller's approval of Buyer's credit.

(A) Machines shipping into US or Canada: 20% deposit with purchase order, 40% prior to shipment, 40% Net 10 Days from Installation Completion. Title to the Goods shall pass to the Buyer only upon receipt of final and full payment of the Goods by Supplier, from Buyer.

(B) Machines shipping into Mexico: 20% deposit with purchase order, 80% paid prior to shipment. Title to the Goods shall pass to the Buyer only upon receipt of final and full payment of the Goods by Supplier, from Buyer.

5. DELIVERY:

The scheduled shipment date is an estimate and is subject to filling prior orders and delays caused by strikes, accidents, shortages, acts of civil authority or other causes beyond Seller's control. Buyer's acceptance of delivery time from the shipper shall constitute a waiver of any claim for delay.

6. INSTALLATION AND PRODUCT ACCEPTANCE:

6A. Preparation of Installation Site:

Upon acceptance by Seller of any Purchase Order hereunder, Seller shall provide Buyer with a Pre-Installation Manual setting out requirements for site preparation at the installation site. Buyer is responsible for preparing the installation site as set out in the Pre-Installation Manual, for providing all utilities within rated parameters as stated in the Pre-Installation Manual, and for inspecting, rigging and placement of the Goods at the installation site.

6B. Inspection Upon Delivery:

Upon arrival of the Goods at the installation site, the Goods shall be inspected for shortages and damage in transit by authorized personnel of both Seller and Buyer. Buyer shall notify Seller in writing of any such claims within ten (10) days after Buyer discovers or should reasonably have discovered facts upon which the claim is based, but in no event more than thirty (30) days after Installation Completion under subsection C below. Failure of the Buyer to give written notice of a claim within the time-period or in the form specified above shall be deemed to be a waiver of such claim. If any repairs are made by unauthorized personnel, Seller reserves the right to withhold warranty support and to charge Buyer per then current time and material rates for services provided if Seller determines that the root cause of a defect requiring service was caused by a third- party vendor supplied product or service.

6C. Acceptance

Testing and Acceptance:

Notwithstanding prior inspection or testing, payment, or passage of title, all Goods shall be subject to final acceptance testing ("Final Acceptance Testing") after installation at the installation site to validate performance against Seller's quoted Technical Specifications. Final Acceptance Testing will be performed by Seller's operator and consist of (i) Seller's standard acceptance-test procedure which checks all aspects of Goods performance against Seller's quoted Technical Specification and (ii) any additional tests that are mutually agreed upon and specified in the purchase order or statement of work referencing these Terms and Conditions. Upon successful completion of the Final Acceptance Testing, the Goods shall be accepted, and Buyer shall sign and deliver to Seller the Seller's acceptance certificate ("Installation Completion").

Notwithstanding the foregoing, the Goods including any software shall be deemed accepted and the written acceptance form certifying Installation Completion shall be deemed executed by Buyer upon the earlier of either of the following two conditions; (x) in the event the Goods are placed into production by Buyer at the Installation Site; or (y) no later than 3 months after the date of delivery as determined by the terms of shipment.

7. EQUIPMENT WARRANTY: Except as otherwise provided in this Section 7, any Goods sold hereunder which (1) at the Installation Site, (2) have been properly installed and maintained by authorized persons, and (3) have been operated within the limits of rated and normal usage, are warranted to conform to Seller's quoted Technical Specifications and to be free of defects in material and workmanship, as determined by Seller's inspection, for a period of one year, or any purchased extended period, coming into effect upon Installation Completion. The terms of this warranty do not apply to any Goods which have a life, under normal usages, that is inherently less than one year. Product demonstrations, test parts, time studies, production estimates and other such particulars furnished to Buyer are only Seller's estimate and do not create a warranty. Within the warranty period, Seller will repair or replace without cost to Buyer any product or parts covered by the warranty which Seller finds to be defective in material or workmanship, provided that the Buyer gives the Seller prompt notice. This shall be the sole and exclusive remedy of the Buyer under this warranty. If Buyer licenses Applications Software to be delivered with Equipment at time of installation, at the end of the Applications Software warranty period Buyer shall receive a free of charge Software Service Agreement for the balance of the Equipment Warranty period.

To facilitate efficient warranty services delivery, Seller may in its sole discretion deliver to the Installation Site a variety of spare parts such as are reasonably believed to be of potential use in completing the Services (such spare parts, together with those parts removed for replacement during Warranty Services, being collectively "Seller's Property.") Buyer will cooperate with Seller in maintaining Seller's Property in a secure location segregated from Buyer's Equipment and Inventory during Services and in providing access during normal business hours for delivery and pick up. Seller is solely responsible for the packaging and cost of shipment of Seller's Property to and from Buyer's Installation Site. Should Buyer elect to retain any of the unused Spare Parts after close of Warranty Services mission, Buyer will be billed for such Parts at a discount of 5% off Seller's List Price as of the date of retention.

Services under this warranty are provided during normal business hours and using standard freight delivery. Services and delivery may be otherwise provided dependent upon availability and will be invoiced at TRUMPF Service rates in accordance with TRUMPF Service policies and practices, which are incorporated by reference herein.

Seller shall not be liable under any warranty or other obligation if the alleged defect in the Goods does not exist or the defect or defective performance is caused in whole or in part by Buyer's or any third party's misuse, neglect, modifications or additions, the use of non-TRUMPF spare parts or consumables, unauthorized attempts to repair, or by accident or other hazard outside Seller's control. Use of a non-TRUMPF spare part shall void the warranty if the TRUMPF spare part is available under the warranty free of charge. The warranty described herein is granted only to the original purchaser of the Goods and original purchaser's specified end-user and is non-transferrable.

DISCLAIMER OF OTHER WARRANTIES: THE WARRANTY DESCRIBED IN THIS SECTION 7 IS IN LIEU OF ALL OTHER WARRANTIES, EXPRESSED OR IMPLIED, INCLUDING, BUT NOT LIMITED TO, ANY IMPLIED WARRANTY OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE EXCLUDED.

8. DATA SERVICES:

Seller's Equipment is self-monitoring and software may collect certain equipment data recording performance metrics of the Equipment during Buyer's use of the Equipment which can be transmitted to Seller in real time via Seller's secure Cloud Connect services portal as set forth in the technical documentation of the Equipment. Cloud Connect services are activated during installation.

Buyers of Equipment utilizing Cloud Connect may receive expanded applications capabilities or enhanced Cloud Connect notification and technician scheduling throughout the Equipment warranty period and during any period in which the Buyer maintains a Seller Service Agreement providing for such notification and reporting services. Seller uses the data collected for delivery of Goods and Services under warranty and service agreements, and in product research and development. Buyer agrees that such data may be transmitted, processed, and used by TRUMPF as set forth herein.

Buyer can opt out of active machine monitoring by following the procedures set forth in the Software and Operator Manuals. Should Buyer exercise this opt out option, it may not receive Cloud Connect enhanced services, may experience limitations on machine functions or systems integrations that rely on information processed through Cloud Connect or automated data communications, or may be subject to surcharges and increased costs.

Data collected via Cloud Connect may include metadata, logs or other information generated by the operation of the Equipment or Embedded Software, in automated logs or similar records through the normal operation of the Equipment. In some cases, and within additional services, NC programs, technology tables, environmental, technology and operation data may also be collected. All data is secured with the same level of care Seller provides for its own confidential information, but in no event less than reasonable care. Some features in Seller software may enable collection of data from users of Buyer's applications that access or use the software. If Buyer uses these features to enable data collection in its application, Buyer is solely responsible for compliance with any applicable law, including obtaining any requisite user consent and maintaining a prominent privacy policy that accurately informs users about how it uses, collects, and shares data.

Seller reserves the right to add and remove services from Cloud Connect offerings at Seller's sole discretion. CLOUD CONNECT SERVICES ARE PROVIDED 'AS IS' WITH NO REPRESENTATION OR WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING WITHOUT LIMITATION: 1) ANY WARRANTY CONCERNING THE AVAILABILITY, ACCURACY OR CONTENT OF THE INFORMATION, OR 2) ANY IMPLIED WARRANTY OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

9. WORKPLACE SAFETY, INDEMNIFICATIONS:

9A. Product Liability:

With respect to all Goods or materials provided under this Agreement that are Seller's standard product or made to Seller's design or specifications, Seller shall, subject to the limitations of liability governing this Agreement, defend and indemnify Buyer against all third-party claims and liability required to be paid by Buyer to the extent arising out of defective materials, workmanship or design by Seller.

- 9B. Infringement/Malware:** Seller shall engage in commercially reasonable efforts to monitor the Goods supplied hereunder for infringement of third-party intellectual property and the presence of viruses or other malicious code. TRUMPF represents that (i) the Goods are not known to infringe, violate or misappropriate any intellectual property right(s) of third parties; (ii) that TRUMPF has all of the rights, permits, licenses and authority known to be necessary to perform its obligations hereunder; and (iii) the Goods, including any software and the media it is delivered on, have been scanned for and been found to be free from viruses and other malicious code. Seller shall, subject to the limitations of liability governing this Agreement, defend and indemnify Buyer against all third-party claims and liability required to be paid by Buyer to the extent arising out of Seller's negligence or malfeasance in monitoring the Goods as provided herein or in making the foregoing representations.
- 9C. Workplace Practices:** It is the responsibility of Buyer to know, understand and comply with the work and safety laws and regulations in effect and governing Buyer's use of the Goods provided hereunder. Buyer agrees that after installation of the Goods provided hereunder, the Goods and all persons other than Seller personnel operating or maintaining such Goods will be deemed under Buyer's exclusive control. To the extent governing law requires inspections, reviews, records keeping and/or after-purchase modifications to the Goods, it is the responsibility of the Buyer to arrange for and comply with such requirements and any associated costs are the sole responsibility of the Buyer. Subject to the limitation of liability governing this Agreement, Buyer shall indemnify Seller against all third-party claims and liability required to be paid by Seller to the extent arising out of (a) any modifications, including but not limited to repairs, made in the Goods by or on behalf of Buyer by persons or parties other than Seller, and/or (b) the Buyer's negligent use of the Goods, including but not limited to use in production with any of the Product safety functions disabled, obstructed or circumvented.
- 9D. Notice; Consent to Settlement:** It is a condition of indemnification or defense under this Section 9 that in the event that any third party asserts a claim or liability with respect to any matter for which a Party is entitled to indemnification hereunder, the indemnified Party gives prompt written notice to the indemnifying Party of such claim or liability. No settlement of an indemnified claim shall require an admission of liability or impose an affirmative obligation on a Party hereto, including an obligation to indemnify, without the written consent of the Party so affected. Consent shall not be unreasonably withheld.
- 9E. Insurance:** Each Party shall maintain comprehensive liability insurance coverage in an amount not less than one million dollars per occurrence with coverage of the other Party as an additional insured. Upon request, a Party shall provide the other Party with a certificate of insurance evidencing this coverage.

9F. Excluded Limitations:

The above indemnifications shall not be limited by the availability of insurance coverage to the indemnifying party, or by any protection afforded the indemnifying party under the Workers' Compensation Acts, Disability Acts or other employee benefits acts.

10. LIMITATION OF LIABILITY:

NEITHER PARTY'S LIABILITY ON ANY INDEMNIFICATION OR ANY CLAIM OF ANY KIND, UNDER ANY THEORY, AT LAW OR IN EQUITY, INCLUDING NEGLIGENCE OR STRICT LIABILITY, FOR ANY LOSS OR DAMAGE ARISING OUT OF, CONNECTED WITH, OR RESULTING FROM THE PERFORMANCE OR BREACH OF THE TERMS HEREOF, OR FROM THE DESIGN, MANUFACTURE, SALE, DELIVERY, RESALE, INSTALLATION, TECHNICAL DIRECTION OF INSTALLATION, INSPECTION, MODIFICATION, REPAIR, OPERATION OR USE OF ANY SERVICE, GOOD OR PART THEREOF SHALL IN ANY CASE EXCEED THE PRICE ALLOCABLE TO THE SERVICE, GOOD OR PART THEREOF WHICH GIVES RISE TO THE CLAIM. IN NO EVENT SHALL EITHER PARTY HAVE ANY LIABILITY FOR ANY INCIDENTAL, CONSEQUENTIAL OR EXEMPLARY DAMAGES ARISING OUT OF OR IN CONNECTION WITH A BREACH OF THE CONTRACT SALE OR ANY OTHER DUTY OF THAT PARTY WITH RESPECT TO THE GOODS OR SERVICES OR THIS AGREEMENT, INCLUDING, BUT NOT LIMITED TO, DAMAGES FOR LOST PROFITS, LOST SALES OR INJURY TO PERSONS OR PROPERTY EVEN IF THE PARTY HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH LOSSES.

11. FORCE MAJEURE:

Except with respect to payment obligations under this Agreement, neither Party shall be liable for or will be considered to be in breach of or default under this contract on account of any delay or failure to perform as a result of any causes or conditions that are beyond the Party's reasonable control and that the Party is unable to overcome through the exercise of commercially reasonable diligence, including but not limited to pandemics, fire, explosion, flood, storm or other acts of God, disruptions in transportation networks, failures of carriers or utilities, war, embargo, strike, riot, or the intervention of any government authority. If any force majeure event occurs, the affected Party shall give prompt written notice to the other party and use commercially reasonable efforts to minimize the impact of the event.

12. LIMITATION OF ACTIONS: No action for breach of any term of this contract of sale or any other duty of Seller with respect to these Goods may be commenced more than one (1) year after Installation Completion.

**13. CHANGES,
CANCELLATION:**

Any changes to the scope of the Goods and Services under this Agreement must be agreed in a writing executed by both Parties. Orders may not be canceled except by written notice received by Seller prior to shipment. A restocking charge of ten percent of the selling price will be applied for the cancellation of standard items. Charges for the cancellation of special items will be based on non-recoverable expenses accruing to the order sustained by Seller plus ten percent of the selling price. Either Party may cancel an existing order without penalty or decline future orders under the Agreement in the event of (a) a change of control of, or (b) a bankruptcy of the other Party.

**14. INTELLECTUAL
PROPERTY RIGHTS:**

The Goods sold hereunder are of Seller's proprietary design offered in configurations and only with such modification as is customarily available to all buyers in the marketplace without significantly altering the Good's, function or processes. The Goods do not incorporate or embody Buyer's intellectual property, or any technology developed by Buyer or specifically for Buyer. Buyer acquires no greater interest in the Goods, or the intellectual property embodied in the Goods than those rights of use, maintenance and resale as customarily accompany the purchase of off-the-shelf tangible goods. All Seller intellectual property including but not limited to patents, trademarks, trade names, trade secrets, and all modifications and derivatives thereof, are and shall remain the sole and exclusive property of Seller.

Any operating software or instruction that is embedded in or incorporated into the hardware or electrical components of the Equipment as an integral part of the Equipment delivered is referred to herein as "Embedded Software." Embedded Software is licensed, not sold. Seller grants Buyer a nonexclusive, perpetual, fully paid, royalty-free restricted license to Embedded Software, in executable form, solely as embedded as part of or through the Equipment and only in conjunction with Buyer's operation of the Equipment and Services as provided by Seller. Buyer may not install, access, configure or use any Embedded Software (including any APIs, tools, databases or other components of any Embedded Software) separately or independently of the rest of the Equipment, whether for production, technical support or any other purposes, or otherwise attempt to gain direct access to any Embedded Software components, or permit anyone else (including Buyer's customers) to do any of these things. Embedded Software excludes applications softwares formulated to interface with end users of the goods to perform specific business or technical functions ("Applications Software") and which are subject to the TRUMPF End User License Agreement.

15. DESIGN CHANGES:

Acceptance of Purchase Order by Seller determines product version for delivery. The designs and specification of all Goods sold may be subject to subsequent change by Seller without notice and, in the event of any such changes, Seller will have no obligation whatsoever to make similar changes in Goods previously ordered.

16. RESERVATION OF SECURITY INTEREST; RISK OF LOSS:

Seller hereby reserves, and Buyer hereby grants a security interest in the Goods (and the proceeds thereof) as security for the payment of the unpaid balance of the purchase price and Buyer's performance of its other obligations hereunder. Buyer will execute and deliver to Seller such financing statements as Seller shall request to perfect such security interest. The security interest hereunder shall terminate upon Buyer's payment in full of the purchase price of Goods and Services, and the respective taxes.

The risk of loss passes to Buyer upon delivery of the Goods to the carrier. Insurance against loss or damage to the Goods during shipment is the responsibility of Buyer. Until the entire purchase price for the Goods is paid in full, Buyer will keep the Goods insured against loss or damage by fire and other risks and hazards included with so-called "extended coverage" insurance, in an amount at least equal to such purchase price. Losses under such insurance shall be made payable to Seller and any payments under such insurance shall be paid to Seller and applied to the unpaid balance of the purchase price. Upon request, Buyer will furnish Seller with copies of the policies of such insurance and each renewal thereof.

17. NONPAYMENT OF PURCHASE PRICE: STORAGE AND COLLECTION COSTS:

If Buyer shall not pay the full purchase price within 30 days from the date of Installation Completion of the Goods; Buyer will pay Seller thereafter an additional one and one-half percent (1-1/2%) per month on the unpaid balance of the purchase price until paid in full. Such charge shall be added to and become an additional part of the purchase price for the Goods. Buyer also will pay all storage costs for the Goods after the scheduled delivery date as well as all costs of collection incurred by Seller in collecting the purchase price for the Goods and enforcing its security interest in the Goods, including, without limitation, reasonable attorneys' fees and expenses incurred by Seller.

18. GOVERNING LAW:

The validity, interpretation and performance of this contract for sale shall be governed by the laws of the State of Connecticut, excluding its conflict of law rules. The United Nations Convention on Contracts for the International Sale of Goods is expressly disclaimed by the Parties with respect to this Agreement and the transactions contemplated hereby.

19. ENTIRE AGREEMENT:

This Agreement constitutes the entire agreement between the Parties pertaining to the subject matter hereof, and supersedes all prior agreements, understandings, negotiations and discussions, whether oral or written, of the Parties pertaining to the subject matter hereof.

Effective Date:

Y2023-01-03 Revisions

Country of origin

The machine plant is manufactured by TRUMPF Maschinen Grösch AG, CH-Grösch/Schweiz. Its customs origin is Switzerland.

Consumables

The lens and nozzle of the lens cutting head as well as the protective glass are subject to wear depending on the particular case and are, like all consumables, not part of the scope of warranty.

Export

This machine is subject to international export control. In accordance with export list position: 6A005For delivery from• the EU according to Article 3, Paragraph 1 of the Regulation (EC) No. 821/2021• Switzerland according to Article 3, Paragraph 1 of the Regulation 946.202.1The delivery is then subject to the granting of an export authorization by the exporting authority responsible for the respective exporting country:• Germany: Federal Office of Economics and Export Control (BAFA)• Switzerland: State Secretariat for Economic Affairs (SECO)In the case of re-export of US products or of US products installed in our machines/plants, a re-export authorization may also be required from the responsible US authority, the Bureau of Industry and Security (BIS).If need be, please contact the contact person responsible for you at TRUMPF.

3. Additional information about the order range

1000 TruLaser 3030 fiber (L95)

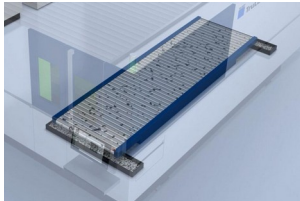
Machine

Box frame

- High level of machine dynamics and accuracy due to high-precision, hardened guides
- Self-stiffening prevents vibrations when changing directions

Longitudinal conveyor belt

- Clean machine interior, reliable removal of slag
- Collection of small parts in easily accessible containers
- No searching for finished parts



Motion unit for high-precision processing

- Optimal processing speed and precision
- Low-weight, temperature-resistant laser-welded cross beam
- Roller-bearing and ball-bearing on flat guides with zero-backlash, rack and pinion drive that acts on both sides

Process cooler for machine and laser (water/air)

- Cools lasers and linear motors in separate circuits
- Low cooling capacity required thanks to energy-efficient TruDisk laser

Work area illumination

- Effective illumination of the entire work area
- Robust lighting design

Pilot laser

- Enables simple, rapid zero point offset in manual operation
- Simple placement of separating cuts
- Good visibility thanks to green laser light

Spraying equipment

- Reduces slag sticking on the upper side of the sheet
- Reduced reworking
- Program-controlled application of spray film on the sheet metal at the piercing spot

Automatic nozzle cleaning

- Longer use of nozzles
- Self-cleaning of nozzles by machine

Integrated electrical cabinets

- Electrical cabinets integrated in the machine frame
- Space-saving, compact installation



Integrated cabinet

- Space for documents, tools and consumables
- Within easy access: directly on the left next to the machine door

Closed beam guidance via laser light cable (LLK)

- Flexible beam guidance from solid-state laser to the cutting unit at the machine
- TRUMPF laser light cable is robust and long-lasting



Machine status light

- Enables color-coded machine status display

Transport and installation device

- Faster start-up of the machine on site due to one-way transport fixture

TRUMPF laser

Excitation using pump diodes

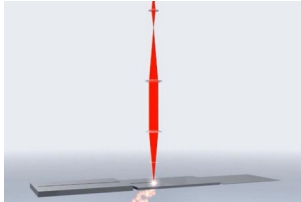
- Very long service life of diodes

Laser power control

- Higher level of parts quality and shorter part times due to dynamically adjusted laser power

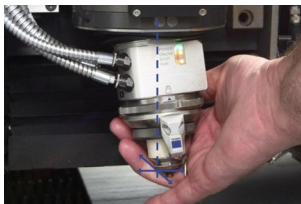
Cutting unit

Single cutting head strategy



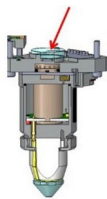
- Universal cutting head enables processing of entire range of sheet thicknesses and materials without replacement of cutting head

Universal cutting unit with adaptive lens system



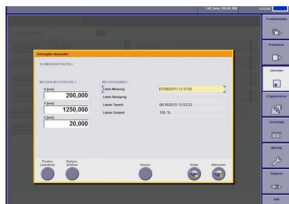
- Highest level of productivity & quality in thin to medium sheet thicknesses and high productivity & quality in thick sheets with a universal cutting unit (variable focus adjustment)
- High level of process reliability and reduction of microjoints thanks to efficient collision protection

Protective glass



- Protects cutting unit lenses from contamination
- Increases process reliability as a result
- Protective glass is less expensive as a consumable than lenses

Online condition checking of protective glass



- Reduced non-productive time and protective glass usage
- Sensor constantly monitors the protective glass condition
- Protective glass cleaned/replaced only as needed

Automatic nozzle centering



- Consistently high cut quality, since the nozzle is automatically centered during loading
- Reduced non-productive time and fewer rejects

Control

Quick post-production

- Reduced programming effort
- Enables reproduction of individual parts from an existing program where required

Ergonomic control panel with 21.5" wide-screen multi-touch color display

- High level of ergonomics due to adjustable control panel
- Fast and efficient operation by intuitive touch interface with simplified user prompts
- Particularly easy, fast-to-learn operation

Integrated technology data

- Wide range of laser technology tables (LTTs) for robust and premium quality processing of commonly available materials
- Program-controlled regulation of process parameters like feed rate and cutting gas pressure

Auto-shutdown

- Electrical energy consumption is reduced in idle state
- Automatic shutdown of the entire machine after the program or job end
- Particularly beneficial in combination with automation

Programmable cutting gas/pressure selection

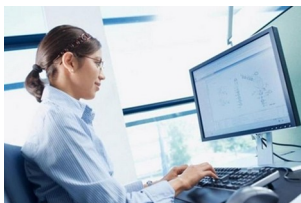
- Program-controlled setting of cutting gas pressure for different materials and material thicknesses
- Display of pressure on operating panel

Microweld

- Small weld point that holds the finished part in the scrap skeleton
- Prevents tilting and falling of finished parts
- Interactive programming in the TruTops programming system

Remote support interface

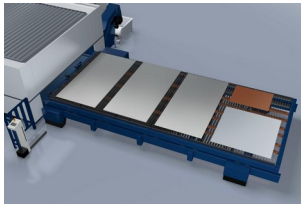
- Contact with Technical Service directly via broadband Internet connection
- Minimization of machine downtimes
- High data security due to passwords
- Requirement: broadband Internet connection available



Production plan

- Several NC programs generated one after the other can be combined in the order of their creation or in the context of a job to form a production package
- Information can be forwarded about the production package to TruTops Fab or a control system (e.g. TC-CELL)
- Unattended execution of NC programs

Multi-sheet processing



- Processing of several sheets on one pallet
- Low level of operator involvement thanks to automatic production plan over multiple sheets
- Can also be used for sheets with different materials
- Optimal utilization of entire work area, even when processing small sheets

Condition Guide

- Higher machine availability due to reduction of unplanned idle states and needs-based maintenance
- Cutting capability of machine can be detected at a glance using traffic light system
- Simple prognosis of required action by means of progression diagram

Cutting Guide

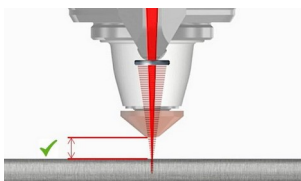
- Helps prevent cutting problems
- Systematic support of operator during setup
- Checks cutting programs for conformity with TRUMPF rules

Functions

AdjustLine

- Increased process reliability when cutting poor material
- Direct access to two alternative cutting data records on machine that increase the robustness of the process

ControllLine



- Capacitive height and process control, maintains distance from nozzle to plate during cutting process and remains constant even with uneven sheets
- Prevents collisions between the cutting head and workpiece

FastLine

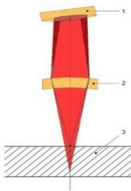
- Optimizes entire cutting process
- Ensures maximum economic efficiency
- Reduces slag spatters on the upper side of the sheet

FlyLine



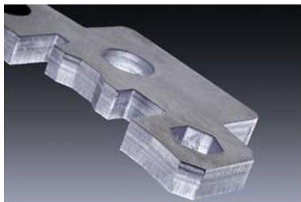
- Flying, positionally-accurate activation/deactivation of laser without axis stop
- Often allows a significant reduction in processing time for parts with many contours

FocusLine



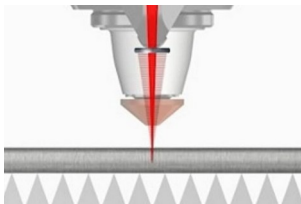
- Enables program-controlled setting of focus position for entire range of materials without manual intervention
- Requirement for maximum processing speed and edge quality

NitroLine



- Nitrogen high-pressure cutting for oxide-free cutting edges
- Can be used for mild steel, stainless steel and aluminum alloys
- Automatic cutting gas control via NC program

PierceLine



- Shorter sheet runtimes thanks to a reduction in piercing times of up to 60%
- Process-reliable piercing even under difficult conditions
- Higher part quality thanks to reduced spatter
- Detection of the end of piercing
- In addition, as of TruDisk 6001: Control and monitoring of piercing process



PlasmaLine

- Higher quality, less wear and improved process reliability when machining thick stainless steel and mild steel thanks to plasma sensor system

Copper cutting package

- Includes cutting data with the cutting gases nitrogen and oxygen
- Stable processing of copper with reliable TRUMPF cutting data

Brass cutting package

- Contains cutting data with the cutting gas nitrogen
- Stable processing of brass with reliable TRUMPF cutting data

Data transmission

USB interface



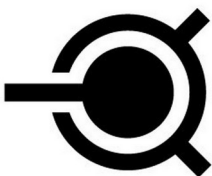
- The machine control is equipped with a USB port for data transmission as standard

RJ45 network interface



- The machine control is equipped with an RJ45 network interface (10 or 100 Mbit/s) as standard

Central Link



- Basic infrastructure for communication with OPC UA interface systems
- Data provided via interface systems (e.g. production status interface)

Online Update Manager

- Higher IT operational safety by the automatic provision of safety updates at the machine
- Better checking of update requirements

Safety

CE marking and EC Machinery Directive

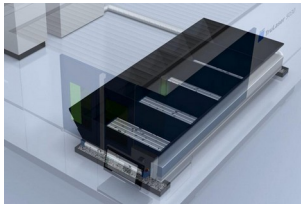


- Delivery with EU declaration of conformity and CE marking
- Machine is subject to the EC Machinery Directive 2006/42/EC, the EMC Directive 2014/30/EU and further directives where relevant
- In individual cases, the machine can be sold only as "partially completed machinery" with declaration of incorporation. Such a machine must be completed by an integrator.

Light barriers

- Protection of working area around the pallet changer by means of multi-beam light barriers

Multi-chamber compact dust extractor

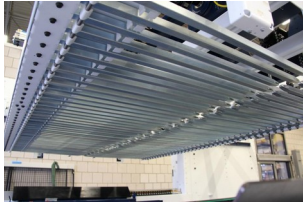


- Efficiently controlled exhaust of fumes and airborne particles
- Compliant with work area limit values for pollutants in the ambient air
- Enables suction into individual chambers so that suction only ever takes place in the chamber located under the area being processed

Enclosed protective housing with certified door viewing panel, right



- No risk of injury due to laser beam
- Optimal accessibility and ergonomics as well as visibility into the machine interior
- One panel included as standard
- Higher productivity and higher maximum sheet thicknesses due to higher laser power
- Low operating costs thanks to high efficiency
- High resistance to back-reflections
- Power stability over service life
- Maximum part quality, even in thick sheet, thanks to BrightLine fiber
- Further increased productivity thanks to fusion-cutting performance package and Highspeed cutting package



- Automated unloading of finished sheets by unloading rake with prongs made of mild steel

- Pallet for sorting cut sheets and finished parts

3000 External solid-state laser TruDisk 6001 with one laser output



- Higher productivity and higher maximum sheet thicknesses due to higher laser power
- Low operating costs thanks to high efficiency
- High resistance to back-reflections
- Power stability over service life
- Maximum part quality, even in thick sheet, thanks to BrightLine fiber
- Further increased productivity thanks to the Highspeed cutting package

4000 External solid-state laser TruDisk 10001 with one laser output



- Higher productivity and higher maximum sheet thicknesses due to higher laser power
- Low operating costs thanks to high efficiency
- High resistance to back-reflections
- Power stability over service life
- Maximum part quality, even in thick sheet, thanks to BrightLine fiber
- Further increased productivity thanks to the Highspeed cutting package

5000 External solid-state laser TruDisk 12001 with one laser output



- Higher productivity and higher maximum sheet thicknesses due to higher laser power
- Low operating costs thanks to high efficiency
- High resistance to back-reflections
- Power stability over service life
- Maximum part quality, even in thick sheet, thanks to BrightLine fiber
- Further increased productivity thanks to the Highspeed cutting package

6000 External solid-state laser TruDisk 24001 with one laser output



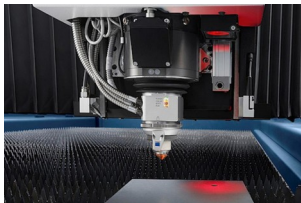
- Higher productivity and higher maximum sheet thicknesses due to higher laser power
- Low operating costs thanks to high efficiency
- High resistance to back-reflections
- Power stability over service life

8000 Nozzle changer



- Higher productivity due to minimal non-productive times: no manual nozzle changes, e.g. with material or process changes
- Low level of operator involvement thanks to fully automatic changes and return of nozzles to storage
- Upon change of process, the nozzle is cleaned with a brush and returned to storage in the nozzle holder; it can be installed again later

9000 DetectLine



- Precise sheet position detection
- Dimensionally accurate and quick further processing of pre-fabricated parts (e.g. punching parts)
- Significant time savings due to comprehensive automation of focal position adjustment

10000 Drop&Cut



- Time savings in the post-production of parts
- Maximize material efficiency by using remainder sheets
- Camera image of remainder sheet is easily filled with part geometries
- Maximum operator convenience and process reliability thanks to cutting preview in live image (augmented reality)
- No programming knowledge required

11000 Integrated camera 1 for process viewing

- Process viewing and rough assessment of cutting result from any location
- Reduction of operator involvement for process monitoring
- Video stream from work area can be shown on external monitor in any location via a DVI interface
- View of the front section of the work area (about 2.5 m)

12000 Integrated camera 2 for process viewing

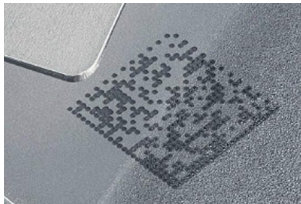
- Process viewing and rough assessment of cutting result from any location
- Reduction of operator involvement for process monitoring
- View of rest of work area (from 6 m format up to 4.5 m)

13000 Smart Collision Prevention / Purchase



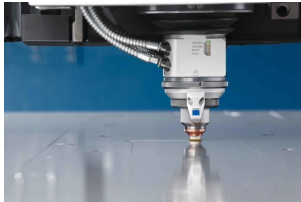
- Innovative execution algorithm prevents collisions between tilting parts and the cutting head
- Sheet layout for the reliable production of parts with a risk of tilting - mostly without microjoints
- Less reworking, quicker part sorting
- Increased material efficiency thanks to closer nesting
- Low level of operator involvement

14000 Dot Matrix Code / Purchase



- Process reliability in sheet metal processing thanks to clear part identification, fewer rejects
- Time savings in the sheet metal process chain thanks to part identification (e.g. activation of correct bending program)
- Productivity through application of codes within seconds during laser processing (minimum interruption of productive time)

15000 Hightspeed Eco cutting package



- Huge gas savings: up to 70% less gas consumption in fusion cutting
- Significantly faster cutting speed: Increase in sheet throughput by up to 100%
- Can be used in fusion cutting of medium sheet thicknesses for mild and stainless steel

16000 CoolLine with water container



- Clear increase in process reliability thanks to targeted water spray in thick mild steel
- Cutting of delicate contours (e.g. thin webs, smaller holes) and of lower quality materials possible
- Material cost savings thanks to closer nesting – up to 25 % less scrap
- The supply from the water container means that no water connection near the machine is necessary

17000 FlexLine

- Significant increase in process reliability in medium to high sheet thicknesses
- Higher material tolerance and lower influence of surface finish on the cutting process
- Improved part quality in various mild steel grades and surface qualities

18000 Compressed-air cutting

- Compressed air can be used as cutting gas (in addition to cutting with oxygen and nitrogen)
- Reduced part costs, in particular with high nitrogen prices
- Use of compressed air affects roughness depth and cutting quality

19000 EdgeLine Bevel / Purchase

- Simple automatic application of chamfers & markings on components
- Various angles and chamfer depths thanks to a wide range of process parameters
- Can be used for mild steel in different sheet thicknesses
- Saves process steps, reduces handling costs and minimizes manual rework

20000 Smart Rerun / Purchase

- Enables the machine to continue production independently after a nozzle collision or cutting flaw occurs without any operator intervention
- Significantly reduces operator involvement, increases productivity time
- Automatic re-entry in the contour; if this is not possible, then jumps to the next part

21000 Titanium cutting package

- Process-reliable cutting of titanium with TRUMPF cutting data
- Processing flexibility: Cutting with argon (shiny edge) and as of TruDisk 4001 also cutting with nitrogen (matte edge) possible

23000 Gas mix cutting package

- Includes cutting data for cutting with a gas mixer
- By cutting with gas mixtures (nitrogen and oxygen) quality improvement (burr formation), but cutting edge oxidation
- Parameters provided with the option must be adjusted by the customer if appropriate.
- In cases of differing material quality, e.g. adjustments to the feed, gas pressure or focus position are necessary

24000 CO2 extinguisher system for compact dust extractor



- Higher level of fire protection by automatic extinguisher system (may be required by fire protection insurance)
- Reduction of downtime

25000 Cutting package for thick sheets

- Contains cutting data for the processing of large sheet thicknesses.
- Only recommended for the production of single parts and small lot sizes.
- Quality (burr formation, roughness) of the parts cut with the option may differ significantly from standard TRUMPF processes.
- Parameters provided with the option must be adjusted by the customer if appropriate.
- For example, adjusting the piercing parameters may be necessary in the event of differing material quality. Using the option may result in the disproportionate consumption of nozzles and/or protective glass.

26000 Clamps (2x for one pallet)



- Optimal part precision and process reliability with the processing of thin sheets or blanks
- No slipping of sheet metal parts during high-pressure cutting
- No tipping of small blanks between the slats
- Holding of warped sheets and pre-processed products

27000 Clamps (4x for two pallets)



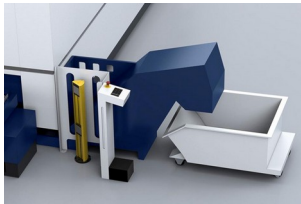
- Optimal part precision and process reliability with the processing of thin sheets or blanks
- No slipping of sheet metal parts during high-pressure cutting
- No tipping of small blanks between the slats
- Holding of warped sheets and pre-processed products

28000 Cleaning brush



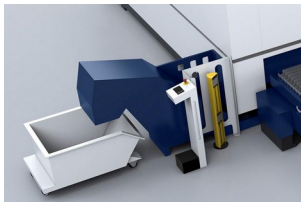
- Removal of loose slag residue or small parts from slats, parallel to pallet change
- Safe processing of subsequent sheets

30000 Transverse conveyor belt with tipping container (to the left of the machine)



- Wide transverse conveyor belt conveys slag and small parts continuously to a storage bin
- Movable container with tilt function

31000 Transverse conveyor belt with tipping container (to the right of the machine)



- Wide transverse conveyor belt conveys slag and small parts continuously to a storage bin
- Movable container with tilt function

33000 MobileControl app



- Transmission of the control panel to the user interface of an iPad
- Flexible operator position for simple monitoring and control of the machine in the immediate vicinity
- App available from the app store

34000 Wireless Operation Point



- Communication between machine and mobile devices (iPad) via a secure wireless network
- Use of MobileControl app (mobile service support) is possible
- Availability of apps and services can be varied depending on machine type

35000 Additional 10-m LLK extension between the laser and machine

- Optimal use of hall layout thanks to flexible setup of TruDisk Laser and machine cooling unit

36000 Right-hand installation of the machine

- Flexible adaptation of machine to spatial and process-related environment
- Units are standing to the right of the machine

37000 Mezzanine version of machine

- Space-saving installation: laser unit, exhaust system and cooling unit on a platform (mezzanine structure) above the machine
- Mezzanine structure is provided by customer, TRUMPF provides corresponding preparation of the machine etc.

38000 Sorting Guide

- Information-supported manual sorting at the pallet changer
- Shorter search times and reduced error rate through user-defined sorting recommendations and automated part identification

39000 Tube cutting unit including four standard and four reversible jaws, support system inside



- Expand the range of potential parts with the addition of tube profiles
- High flexibility in tube geometry (diameter and side ratio)
- No setup work required on cutting unit (no adapter)
- Supports move into the machine automatically during processing

40000 External support max. D 20-165 mm (for pipes up to 6 m, includes two step rollers)

- Fast tube guidance for small part production by means of movable step roller
- Shielding of laser light from the machine
- The support can be quickly uncoupled, if needed

41000 Additional internal support D 20-370 mm

- Improvement in the workpiece precision during continuous processing
- Supports move into the machine automatically during processing
- Support for tubes with outer circle diameter 20-370 mm

43000 LoadMaster 1530

Loading

Loading station

- Provision and alignment of raw sheets

44000 Peeling unit

- Separates sheets that are stuck together when lifting the top sheet
- Peeler suction cup with special stripping insert, for sheet thicknesses up to 8 mm
- Greater process reliability during loading due to clean cutting of the sheets

45000 Second light barrier circuit



- Optical safety sensors for safeguarding the danger zone

46000 Front dual loading cart B1

- Double cart system for provision of raw material
- Increase in raw material capacity
- Increases productivity due to loading outside the safety circuit
- Reduced setup times
- Greater flexibility due to two raw material places

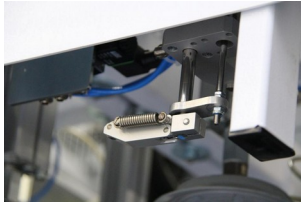
47000 Front loading cart B1

- Cart system for provision of raw material
- Increases productivity due to loading outside the safety circuit

49000 LiftMaster 1530 (AL35)

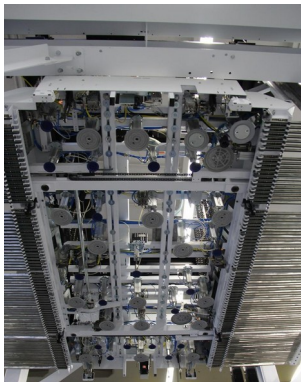
Loading

Double sheet detection



- Detects the thickness of the lifted sheet using a probe arm
- Detection of sheets that are stuck together
- Greater process reliability when loading sheets

Suction frame

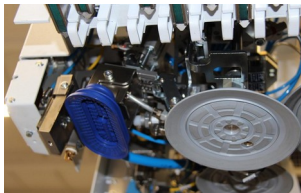


- Automatic loading of raw material by individually controllable suction cups
- Reduced machine waiting times for loading automation

Peeling unit

- Separates sheets that are stuck together when lifting the top sheet
- Peeler suction cup with special stripping insert, for sheet thicknesses up to 8 mm
- Greater process reliability during loading due to clean cutting of the sheets

Separation of thin sheet



- Separation of thin sheets with a thickness of 0.8–2.0 mm is supported by a special peeler suction cup, fanning magnet and air nozzle
- Separates sheets that are stuck together, especially film-coated or oiled sheets
- Greater process reliability when loading thin sheets

50000 Loading station

- Provision and alignment of raw sheets

51000 Unloading platform

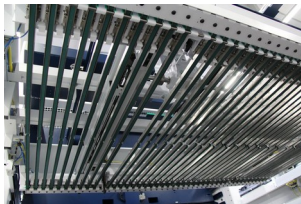
- Unloading station for cut sheets

52000 Rake with belt prongs for low-scratch, positioning accurate unloading (prong spacing 67 mm)



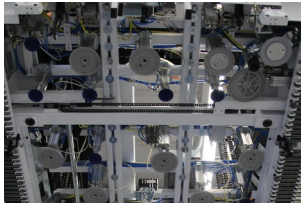
- Automated unloading of finished sheets by unloading rake with belt prongs
- Enables protective, low-scratch and precisely positioned unloading of the sheets
- For sheets up to a sheet thickness of 15 mm

53000 Rake with belt prongs for low-scratch, position accurate unloading (prong spacing 33.5 mm)



- Automated unloading of finished sheets by unloading rake with belt prongs
- Enables protective, low-scratch and precisely positioned unloading of the sheets
- For sheets up to a sheet thickness of 15 mm

54000 Part separation from 600 × 600 mm



- Automatic separation of parts and scrap skeletons
- Additional, individually controllable suction cups separate finished parts from a size of 600 × 600 mm from the scrap skeleton.
- During unloading, the laser scanner checks whether all finished parts have been completely separated from the scrap skeleton.
- Reduction of the manual sorting time
- Stacked parts ready for further processing

55000 Integrated sheet alignment



- Device for alignment of raw sheet on the rake frame
- Parallel-to-production alignment of sheets during transport
- Increased process reliability for part separation

57000 Base for LiftMaster

- Layout-dependent increase for connection to storage systems

58000 Storage interface

- Allows stores to be connected to the machine or the loading/unloading system
- Definition of data exchange between a TRUMPF machine and a store for raw material or finished parts

59000 LiftMaster Compact 1530 (AL34)

Basic unit

Two gantry stations



- Station for loading and unloading or for machine connection

Extended pallet removal unit

- Greater distance between machine and pallet changer
- Frequently required if the machine is operated with automation

Storage interface

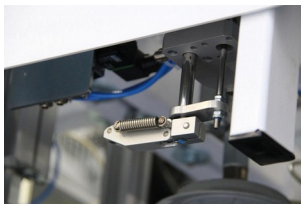
- Allows stores to be connected to the machine or the loading/unloading system
- Definition of data exchange between a TRUMPF machine and a store for raw material or finished parts

Touch start post

- For manual operation of the automation
- Allows error messages to be displayed in another location
- The button functions are indicated with a pictogram and a mark
- If the system is later expanded, a software update is sufficient to adapt the operating elements

Loading

Double sheet detection



- Detects the thickness of the lifted sheet using a probe arm
- Detection of sheets that are stuck together
- Greater process reliability when loading sheets



Synchronous loader including separation of thin sheet

- Parallel loading of raw material on the pallet changer and storage of processed sheets on the finished part pallet of the synchronous loader
- Enables fast loading and unloading
- Separation of thin sheets with a thickness of 0.8 – 2.0 mm is supported by a special peeler suction cup and nozzle
- Greater process reliability when loading thin sheets

Peeling unit

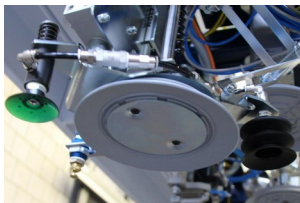
- Separates sheets that are stuck together when lifting the top sheet
- Peeler suction cup with special stripping insert, for sheet thicknesses up to 8 mm
- Greater process reliability during loading due to clean cutting of the sheets

Cleaning brush for slats

- Removes loose slag residue or small parts from the slats
- Increases the process reliability for the processing of the subsequent sheets



60000 Thin sheet handling



- Enables gentle loading of sheets from a 0.4 mm sheet thickness
- Suction cup inserts prevent deformation of the sheet
- Double lifting cylinder on each suction cup to move to different end positions
- This makes it possible to put down the sheets gently

61000 Load station for expanded stack height



- Loading station with stroke function
- Expands the maximum stack height for raw material up to 300 mm

62000 Loading station for expanded stack height in TruStore 3030

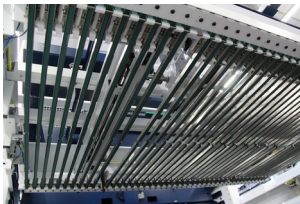


- Expands the maximum stack height for raw material up to 170 mm

63000 Unloading platform

- Unloading station for cut sheets

64000 Rake with belt prongs for low-scratch, positioning accurate unloading (prong spacing 67 mm)



- Automated unloading of finished sheets by unloading rake with belt prongs
- Enables protective, low-scratch and precisely positioned unloading of the sheets
- For sheets up to a sheet thickness of 15 mm

65000 Rake with belt prongs for low-scratch, position accurate unloading (prong spacing 33.5 mm)



- Automated unloading of finished sheets by unloading rake with belt prongs
- Enables protective, low-scratch and precisely positioned unloading of the sheets
- For sheets up to a sheet thickness of 15 mm

66000 PartMaster



- Fast, ergonomic sorting of finished parts and scrap skeletons
- Suitable for sheet thicknesses up to 12 mm

67000 PartMaster at a gantry station



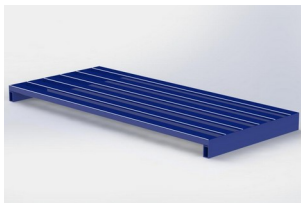
- Fast, ergonomic sorting of finished parts and scrap skeletons
- Suitable for sheet thicknesses up to 12 mm

68000 PartMaster integration TruStore



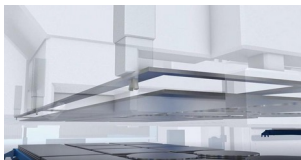
- Fast, ergonomic sorting of finished parts and scrap skeletons
- Suitable for sheet thicknesses up to 12 mm
- Additional installation of up to five storage locations possible
- Compact, space-saving layout

69000 Additional finished part pallet



- Additional pallet for storing finished parts
- Replacement of finished part pallet using a forklift truck

70000 Part separation from 150 × 150 mm



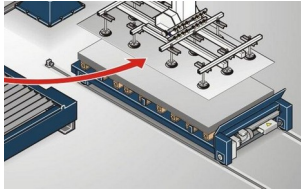
- Automatic separation of parts and remainder sheet:
Expansion grippers hold remainder sheet, finished parts on closed unloading rake ready for stacking
- Separation of parts from a size of 150 × 150 mm up to a sheet thickness of 6 mm
- Reduction of the manual sorting time
- Stacked parts ready for further processing

71000 Linear version



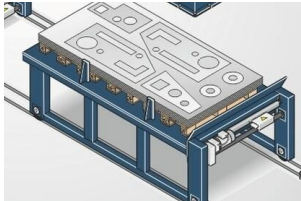
- Offers higher flexibility for the material flow
- Can be combined with cart systems, PartMaster or unloading platform
- Improved access to finished parts

73000 Loading cart



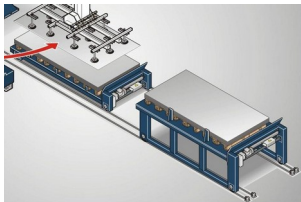
- Enables partially automated provision of raw materials
- Reduces material handling times
- Available in a wide range of formats

74000 Unloading cart



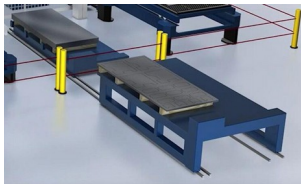
- Enables semi-automated removal of finished parts from the system
- Reduces material handling times
- Available in a wide range of formats

75000 Dual loading cart



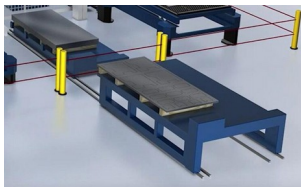
- Increase in raw material capacity
- Supply with two different material types
- Parallel-to-production, manual loading of carts
- Reduced setup times
- Semi-automatic connection to store possible

76000 Double unloading cart



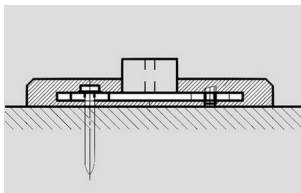
- Increase in unloading capacity for finished parts
- Parallel to production, manual unloading of carts
- Semi-automatic connection to store possible

77000 Double loading / unloading cart



- Increase in raw material and unloading capacity for finished parts
- Supply with two different material types
- Material supply for the machine and unloading of the finished parts mostly parallel to production
- Semi-automatic connection to store possible

78000 Travel rails, surface-mounted

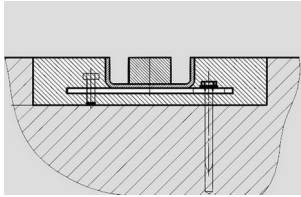


- Travel rails for cart systems
- Adaptation of travel rails to local conditions

Technical Quotation

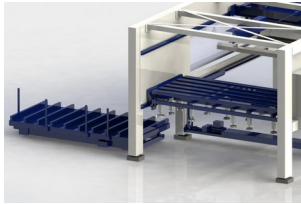


79000 Travel rails, embedded



- Travel rails for cart systems
- Adaptation of travel rails to local conditions

80000 Front loading cart (with scissor lift) for expanded stack height



- Cart system for provision of raw material
- Improved accessibility for loading operation

96000 Slim Nozzles Add-on

5 packs of any size EAB nozzles

97000 BrightLine Add-on

5 pieces of any size BrightLine nozzles

98000 Performance and Highspeed Nozzles Add-on

3 packs of any size Performance and High Speed nozzles

99000 CoolLine Add-on

7 pieces of any size CoolLine nozzles & 1 universal ceramic

100000 Highspeed Eco Add-On

1 High Speed Eco EAV nozzle
2 packs of EAV inserts
2 packs of EAV sleeves

4. Additional information on technical data

TruLaser 3030 fiber (L95)

Control, hardware, electrics

Control type	Siemens SINUMERIK
Operating system	Windows 10 IoT
Screen	multi-touch screen
Screen size	21.5 in
Interface	USB

Axis data - max. axis speed

Max. axis speed simultaneous	170 m/min
Axis range X axis	120 m/min
Max. axis speed Y axis	120 m/min
Z axis	80 m/min

Axis data - Max. axis speed (marketing values)

Max. axis speed simultaneous	170 m/min
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Axis data - Accuracy

Accuracy central positioning variation Ps	0.03 mm
Accuracy positioning deviation Pa	0.05 mm
Accuracy smallest programmable increment	0.001 mm

Axis data - Max. axis acceleration, cutting

Axis range X axis	7 m/s ²
Max. axis speed Y axis	7 1/min

Axis data - Max. axis acceleration, positioning

Axis range X axis	10 m/s ²
Max. axis speed Y axis	17 m/s ²

Technical Quotation



Working range, workpiece size

Axis range X axis	3000 mm
Max. axis speed Y axis	1500 mm
Z axis	116 mm
Max. workpiece weight up to TruDisk 6001	1100 kg
Max. workpiece weight from TruDisk 8001 on a pallet (with heavy-duty pallet changer)	1800 kg
Max. workpiece weight from TruDisk 8001 on both pallets in total (with heavy-duty pallet changer)	2200 kg

Laser-specific data - TruDisk 4001 ¹

Max. laser power	4000 W
Max. sheet thickness, stainless steel	20 mm
Max. sheet thickness, aluminium	20 mm
Max. sheet thickness, copper	8 mm
Max. sheet thickness, brass	8 mm
Max. mild steel sheet thickness with BrightLine fibre	22 mm
Max. sheet thickness, mild steel ²	25 mm
Max. sheet thickness, stainless steel ³	35 mm

Laser-specific data - TruDisk 6001 ⁴

Max. laser power	6000 W
Max. sheet thickness, stainless steel	25 mm
Max. sheet thickness, copper	10 mm
Max. sheet thickness, brass	10 mm
Max. mild steel sheet thickness with BrightLine fibre	25 mm
Max. sheet thickness, mild steel ⁵	32 mm
Max. sheet thickness, stainless steel ⁶	35 mm
Max. aluminium sheet thickness with BrightLine fibre	25 mm

Technical Quotation



Laser-specific data - TruDisk 8001 ⁷

Max. laser power	8000 W
Max. sheet thickness, copper	12.7 mm
Max. sheet thickness, brass	10 mm
Max. mild steel sheet thickness with BrightLine fibre	25 mm
Max. sheet thickness, mild steel ⁸	32 mm
Max. stainless steel sheet thickness with BrightLine fibre	30 mm
Max. sheet thickness, stainless steel, with cutting package for thick sheets	35 mm
Max. aluminium sheet thickness with BrightLine fibre	25 mm

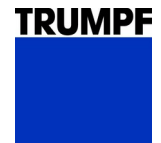
Laser-specific data - TruDisk 10001 ⁹

Max. laser power	10000 W
Max. sheet thickness, mild steel	30 mm
Max. sheet thickness, aluminium	30 mm
Max. sheet thickness, copper	12.7 mm
Max. sheet thickness, brass	12.7 mm
Max. sheet thickness, mild steel ¹⁰	32 mm
Max. stainless steel sheet thickness with BrightLine fibre	40 mm

Laser-specific data - TruDisk 12001 ¹¹

Max. laser power	12000 W
Max. sheet thickness, mild steel	30 mm
Max. sheet thickness, aluminium	30 mm
Max. sheet thickness, copper	12.7 mm
Max. sheet thickness, brass	12.7 mm
Max. sheet thickness, mild steel	35 mm
Max. stainless steel sheet thickness with BrightLine fibre	40 mm
Max. sheet thickness, stainless steel	50 mm

Technical Quotation



Laser-specific data - TruDisk 24001

Max. laser power	24000 W
Max. sheet thickness, mild steel, O2	40 mm
Max. sheet thickness, mild steel, with cutting package for thick sheets	60 mm
Max. stainless steel sheet thickness with BrightLine fibre	40 mm
Max. sheet thickness, stainless steel, with cutting package for thick sheets	60 mm
Max. aluminium sheet thickness with BrightLine fibre	40 mm
Max. sheet thickness, copper, O2	16 mm
Max. sheet thickness, brass, N2	12.7 mm

Consumption values - TruDisk 4001

Average power input in production	13 kW
Average Compressed air consumption	5 m³/h

Consumption values - TruDisk 6001

Average power input in production	15 kW
Average Compressed air consumption	5 m³/h

Consumption values - TruDisk 8001

Average power input in production	17 kW
Average Compressed air consumption	5 m³/h

Consumption values - TruDisk 10001

Average power input in production	19 kW
Average Compressed air consumption	5 m³/h

Consumption values - TruDisk 12001

Average power input in production	25 kW
Average Compressed air consumption	5 m³/h

Consumption values - TruDisk 24001

Average power input in production	33 kW
Average Compressed air consumption	5 m³/h

Paint

Main colour	white, NCS S 0505-R80B
Secondary colour	blue, RAL DESIGN 250 20 20
Switch cabinet/periphery	gray, RAL 7040

Technical Quotation



RotoLas

Min. tube diameter or outer circle	20 mm
Max. effective aperture of chuck	165 mm
Max. positioning deviation	0.03 °
Backlash	0.01 °

LoadMaster 1530

Loading

Max. sheet size (W)	3048 mm x 1524 mm
Min. sheet size (L)	1000 mm
Min. sheet size (W)	500 mm
Max. sheet thickness	25 mm
Max. sheet weight	900 kg

LiftMaster 1530 (AL35)

Loading

Max. sheet size (L)	3048 mm
Max. sheet size (W)	1524 mm
Min. sheet size (L)	1000 µm
Min. sheet size (W)	1000 µm
Max. sheet thickness	25 mm
Max. sheet weight	900 kg

Unloading

Max. sheet size (L)	3048 mm
Max. sheet size (W)	1524 mm
Min. sheet size (L)	150 mm
Min. sheet size (W)	150 mm
Max. sheet weight	900 kg
Max. sheet thickness with unloading rakes	25 mm
Max. sheet thickness with belt prongs	15 mm
Max. sheet thickness with auxiliary pallets	25 mm
Min. part size (L)	150 mm
Min. part size (W)	150 mm

LiftMaster Compact 1530 (AL34)

Technical Quotation



Loading

Max. sheet size (L)	3048 mm
Max. sheet size (W)	1524 mm
Min. sheet size (L)	1000 mm
Min. sheet size (W)	1000 mm
Max. sheet thickness	25 mm
Max. sheet weight	900 kg

Unloading

Max. sheet size (L)	3048 mm
Max. sheet size (W)	1524 mm
Min. sheet size (L)	150 mm
Min. sheet size (W)	150 mm
Max. sheet weight	900 kg
Max. sheet thickness with unloading rakes	25 mm
Max. sheet thickness with belt prongs	15 mm
Max. sheet thickness with auxiliary pallets	25 mm

^{1, 4, 7, 9, 11} This is a selection of sheet thickness specifications with TRUMPF cutting data, detailed information in the data collection on request

^{2, 3, 5, 6, 8, 10} With cutting package for thick sheets option